

Airline Reservation System

A terminal-based Airline Reservation System developed in C++ that allows users to manage flights and passenger bookings efficiently. The project demonstrates the implementation of fundamental Data Structures and Algorithms, including arrays, linked lists, searching algorithms, and sorting algorithms.

Features

Flight Management

- Add new flights
- Update flight information
- Delete flights
- Display all available flights
- Search flights using:
 - Linear Search
 - Binary Search
- Sort flights using:
 - Bubble Sort (by Flight Number)
 - Quick Sort (by Departure Time)

Booking Management

- Book flight tickets
- Cancel bookings
- Update passenger information
- Search bookings
- Display all bookings

Technologies Used

- C++
- Object-Oriented Programming (OOP)
- Arrays (using Vector)
- Linked Lists
- Searching Algorithms
- Sorting Algorithms

Data Structures Implemented

Data Structure	Purpose
Vector (Array)	Store flight records
Linked List	Manage booking records

Algorithms Used

Searching

- Linear Search
- Binary Search

Sorting

- Bubble Sort
- Quick Sort

Learning Outcomes

This project demonstrates:

- Object-Oriented Programming concepts
- Dynamic memory management
- Data structure implementation
- Searching and sorting algorithm analysis
- Console-based application development

How to Run

1. Clone the repository:

```
git clone https://github.com/abdulrehman2301/Airline-Reservation-System.git
```

2. Navigate to the project directory:

```
cd Airline-Reservation-System
```

3. Compile the source code:

```
g++ main.cpp -o airline
```

4. Run the executable:

```
./airline
```

Project Structure

Airline-Reservation-System/

```
|  
├── main.cpp  
├── README.md  
├── Project_Report.pdf  
└── Screenshots/
```

Contributors

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License

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